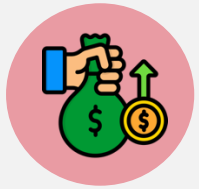
Popcorn & Predictions: What Makes a Movie a Hit?

Team Project

Business Goals



1. Predict Movie Profitability

Build a supervised learning model to estimate a film's profitability based on features like budget, genre, popularity, and ratings.

2. Analyze Success Factors

Identify which movie attributes most strongly influence financial performance and audience popularity

3. Strategic Movie Segments

Use clustering to group similar movies and uncover patterns that support targeted marketing and production strategies

Business Impacts

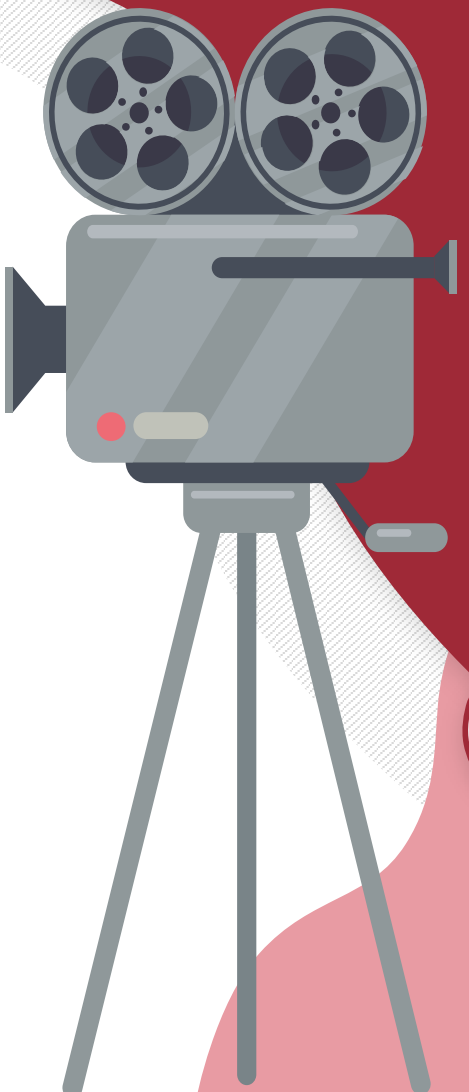
Reduce financial risk by predicting profitability before production

Support data-driven decisions for studios, investors, and streaming platforms

Optimize budgets & marketing using insights from success factors

Research Questions

1. Can we build a model to predict profitability category using only pre-release features?
2. What production and release features most strongly influence the magnitude of profitability?
3. Are there distinct clusters of movies that share similar success profiles?
4. How do genre combinations and release years affect audience popularity trends?
5. Can we identify underperforming movies with high budgets but low profitability, and what patterns do they share?



Data Set and Preprocessing

Overview

Data Size

- 10,000 Popular movies
- 15 Features

Key Variables

- Budget
- Revenue
- Popularity
- Genres

Target Variable

- **Revenue** (used in supervised learning)

Data Challenges

- Missing Values
- Zero Budgets
- Nested genre lists

Data Cleaning

1. Narrowed Dataset Scope

Filtered movies by vote count, budget, release year, and data completeness to create a high-quality, modern subset for analysis.

2. Processed Multi-Valued Columns

Converted genre and production company lists into binary features

3. Removed Pre-Release Films

Excluded movies without a final release date to ensure all ratings, revenue, and popularity metrics are valid.

4. Defined Popularity & Rating Metrics

Created consistent popularity, engagement, and rating categories using vote count, vote average, and popularity scores.

5. Extracted Date-Based Features

Used the datetime extraction function to generate features like year, month, quarter, season, and release timing indicators.

Model- Unsupervised Learning

Model Introduction

Clustering K-means

1 Financial Performance

Helps distinguish major blockbusters, indie films, and underperformers based on financial outcomes

2 Audience Engagement

Captures how well a movie resonates with audiences in terms of quality perception and engagement

3 Release Timing

Reflects seasonal and long-term trends that influence movie performance

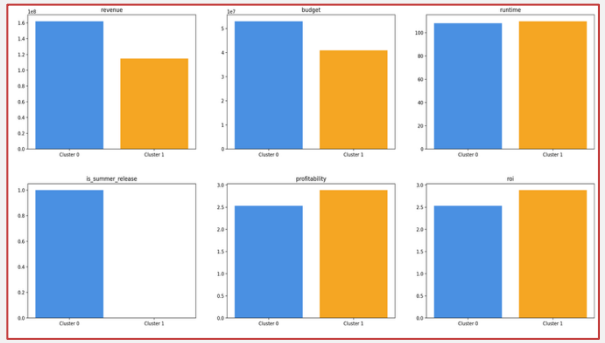
4 Other Runtime

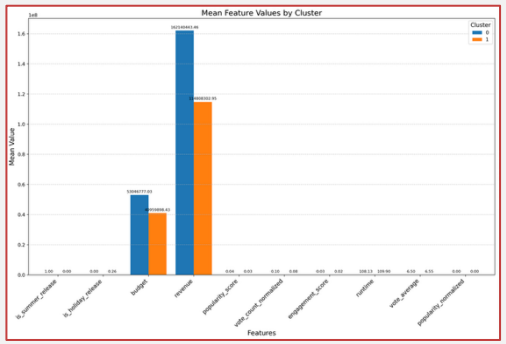
Provides additional context on movie type and production style

Model Result

Cluster 0 → High-budget, high-revenue films, mostly summer releases, with stronger audience

Cluster 1 → Lower-budget films with modest revenue and smaller audiences





Unsupervised Learning- Patterns Discovered

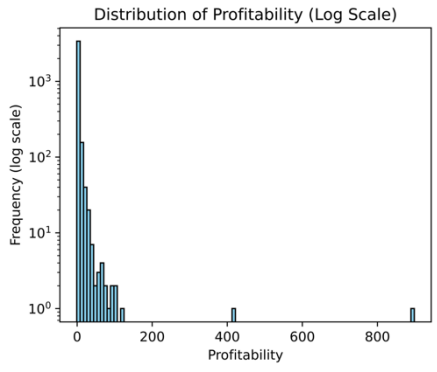
	Clustering	Performance	Strategic Insights
Clustering 0	• High Investment, Large Productions. • Release during high-demand periods (peak in the summer)	• Consistently higher box-office revenue. • Stronger audience engagement and sustained demand.	• Expand global distribution footprint. • Prioritize investment in blockbuster-scale projects. • Optimize release timing around industry peak periods to maximize ROI.
Clustering 1	• Lower-budget, smaller-scale films. • Lower production and marketing expenditure	• Moderate revenue generation. • More limited audience reach and brand visibility.	• Focus on niche audience segments with tailored content. • Maintain strict cost efficiency to secure profitability. • Use targeted marketing rather than mass-market campaigns

Model- Supervised Learning

Model Introduction

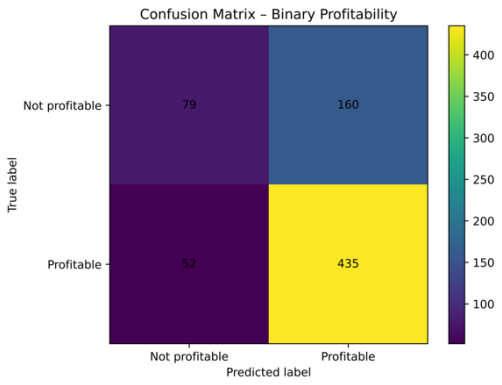
①. Random Forest Regression

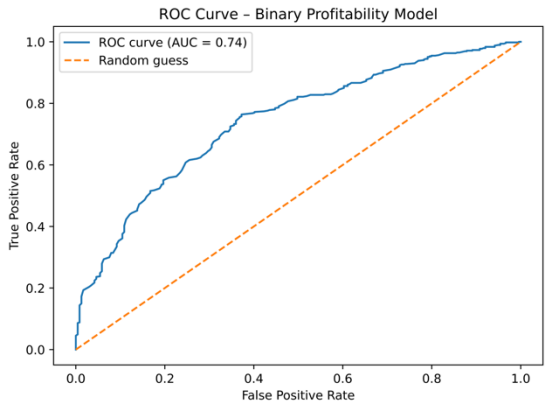
- Predicts the continuous profitability value of a film.
- Helps identify key drivers influencing how much a movie earns



②. Random Forest Multi-Class Classification

- Predicts the profitability tier (major loss → highly profitable).
- Evaluates whether pre-release features can distinguish different financial outcomes.

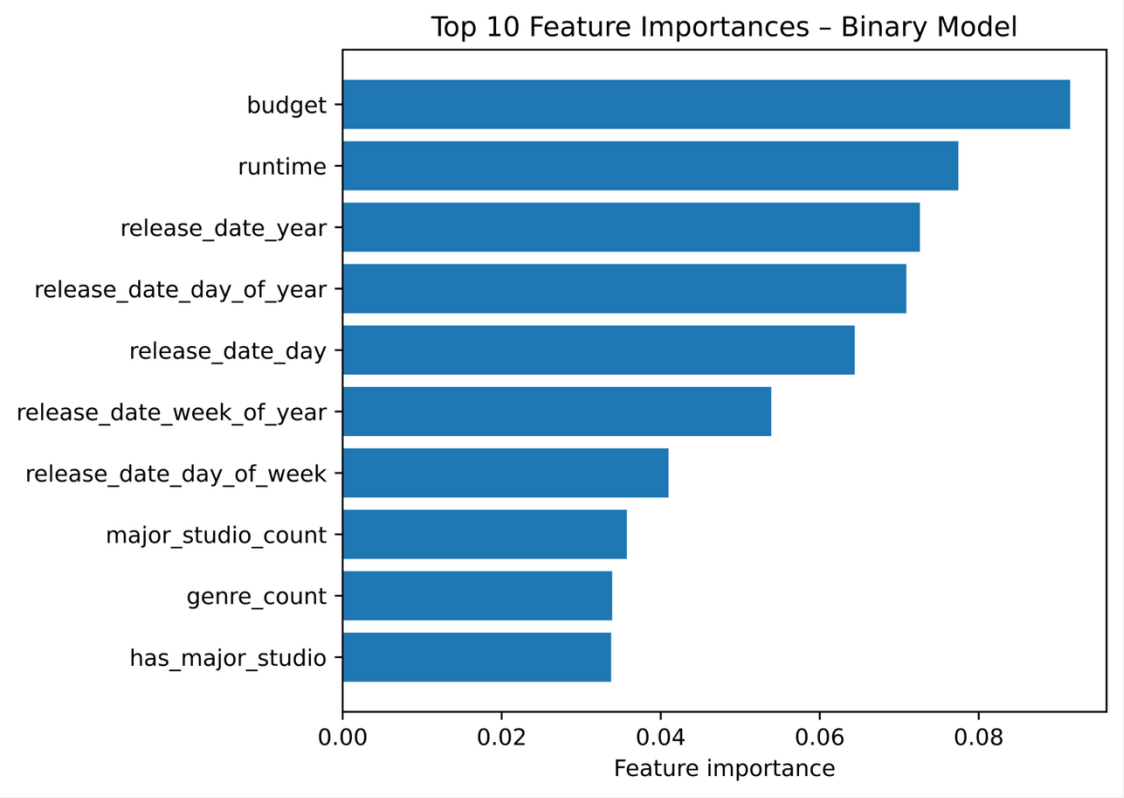




Model Result

③. Random Forest Binary Classification

- Predicts whether a film will be profitable or not.
- This model achieved the best performance (AUC = 0.74) and offers strong business value



Limitations

Unsupervised Learning:

1 Categorical Variables Require Encoding

- × Only works well predominantly with numerical data.
- × Categorical data must be encoded (e.g., genres, studio names), which may distort distances.

2 Manual and Subjective Feature Selection

- × K-means requires manual feature selection, and the clustering outcome heavily depends on which variables are chosen.

3 Sensitivity to Scaling and Distance Assumptions

- × K-means relies heavily on distance-based similarity, meaning results can shift significantly depending on how variables are scaled or standardized.

Supervised Learning:

1 Outliers and Excluded Data

- × Profitability is extremely skewed and dominated by a few outliers
- × Important post-release factors like marketing, competition, and audience reception are not included, limiting predictive accuracy.

Business Limitations:

- × Unexpected events (competition, seasonality shifts, economic conditions) can significantly affect outcomes but are not modelled.
- × Clusters based on historical data may not hold for new genres, streaming-first releases, or shifting consumer preferences
- × Profitability is influenced by many real-world factors that we cannot observe, such as marketing intensity and market competition, limiting the model's practical accuracy. The model should therefore be viewed as a decision-support tool rather than a full replacement for studio judgment

Recommendations

Insights	Recommended Actions
Prioritize Higher-Budget Films *Budget emerged as a strong profitability predictor*	• Invest in high-quality, higher-budget productions that consistently deliver stronger commercial returns. • Avoid lowering budgets to cut costs, as this may reduce production value and limit revenue potential.
Release Films during High-Demand Period *Release timing (year, week, day) was highly influential*	• Schedule film launches during peak demand periods (e.g., summer holidays & major events) to maximize audience turnout. • Align release timing with genre themes (e.g., thrillers in October, family films during holidays) to amplify marketing impact.
Keep Runtime Within an Optimal Range *Runtime ranked second in feature importance*	• Maintain runtimes that fit genre norms to keep audiences engaged and increase showtime frequency per day. • Avoid extremes in runtime, as both overly long and overly short cuts negatively affect viewer experience and revenue.

Applications and Governance

Use cases

- Enhance personalized movie recommendations using audience engagement metrics
- Identify optimal release windows based on historical success, competition intensity, and thematic relevance
- Guide streaming platforms in movie acquisition by predicting profitability for films with strong long-term engagement potential and improved content ROI
- Help marketing teams design targeted campaigns using clustering insights to customize strategies, large-scale campaigns for high-budget films and niche targeting for smaller productions
- Deploy predictive models to anticipate audience demand across different time periods and inform data-driven release decisions

Digital Governance

- Ensure responsible use of ML predictions to avoid reinforcing bias toward high-budget or mainstream films
- Secure usage of streaming and production insights by storing internal data securely and restricting access to authorized decision-makers only
- Update the model regularly to reflect changes in audience trends and market dynamics
- Create governance frameworks for model deployment that include human oversight in critical business decisions
- Monitor model performance metrics continuously to detect drift, degradation, or unintended consequences in predictions

Exhibits - EDA

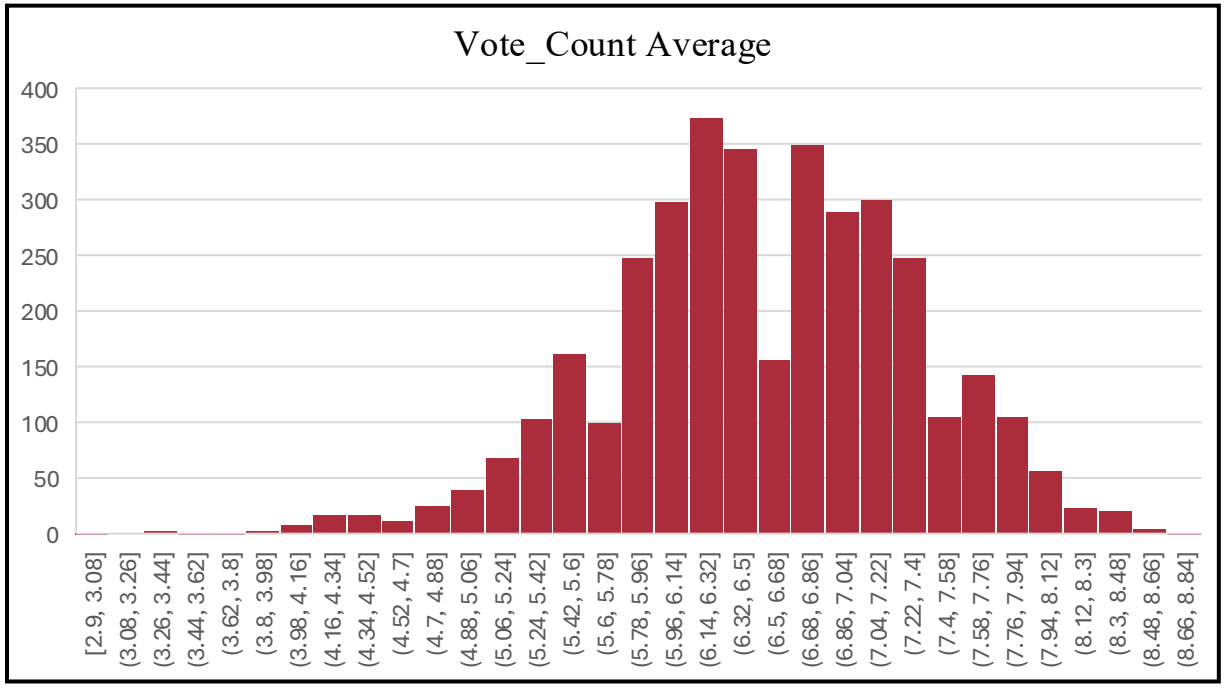


Fig 1. Distribution of Average User Ratings

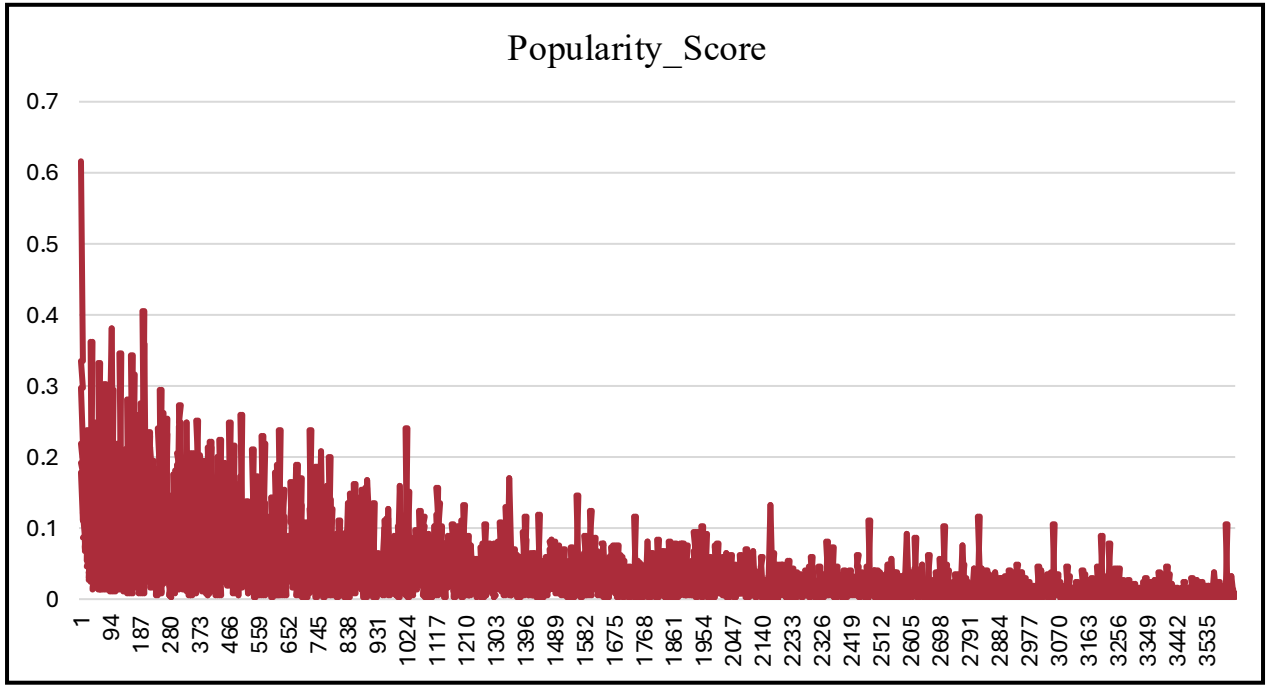


Fig 2. Distribution of Normalized Popularity Scores (Right-Skewed)

Exhibits - EDA

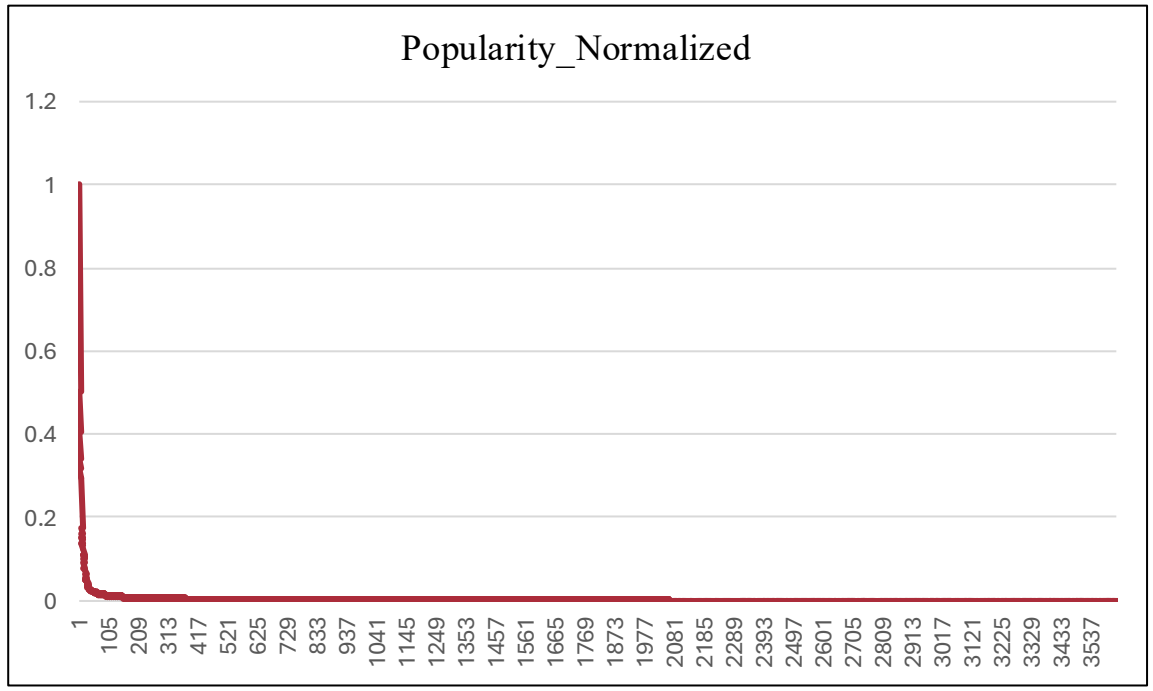


Fig 3. Normalized Popularity Distribution

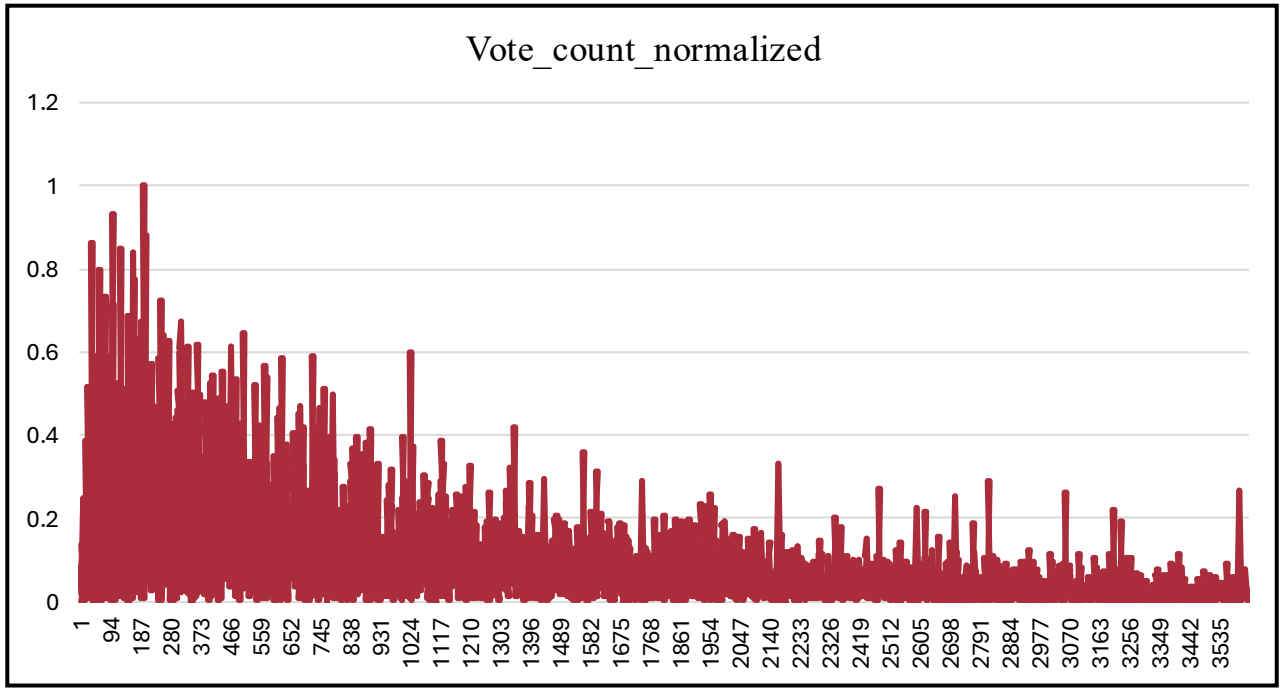


Fig 4. Normalized Vote Count Distribution (Right-Skewed)

Exhibits - EDA

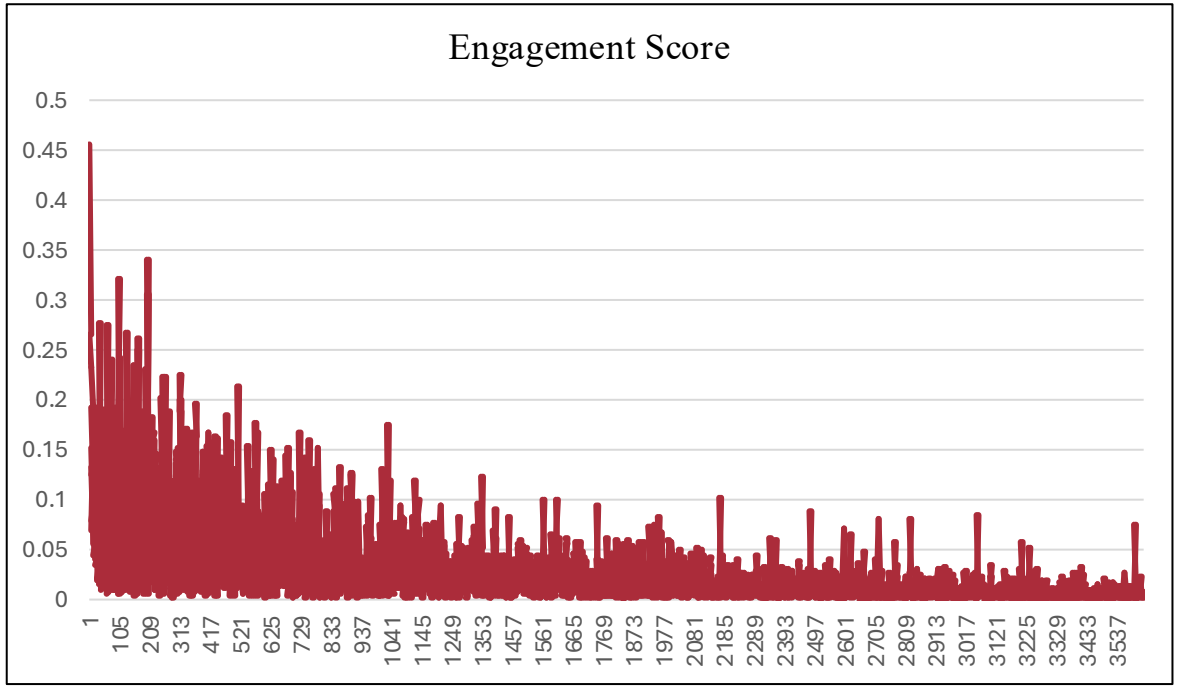


Fig 5. Composite Engagement Score Distribution

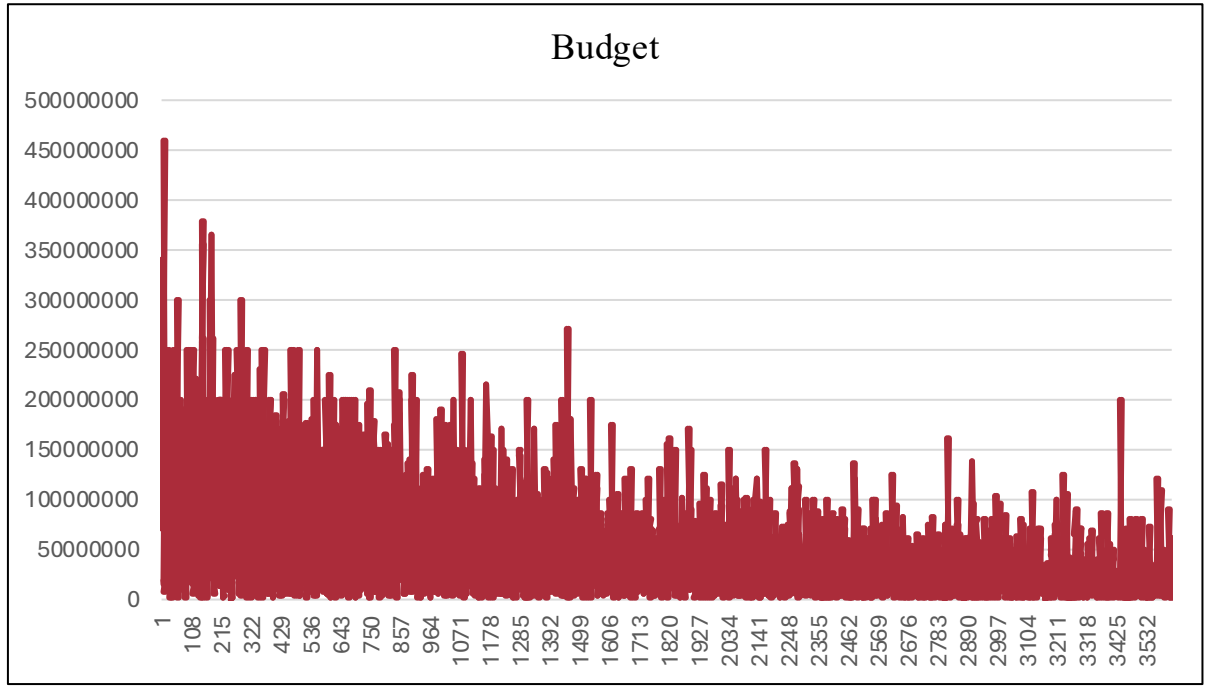


Fig 6. Composite Budget Distribution

Exhibits – Supervised Learning

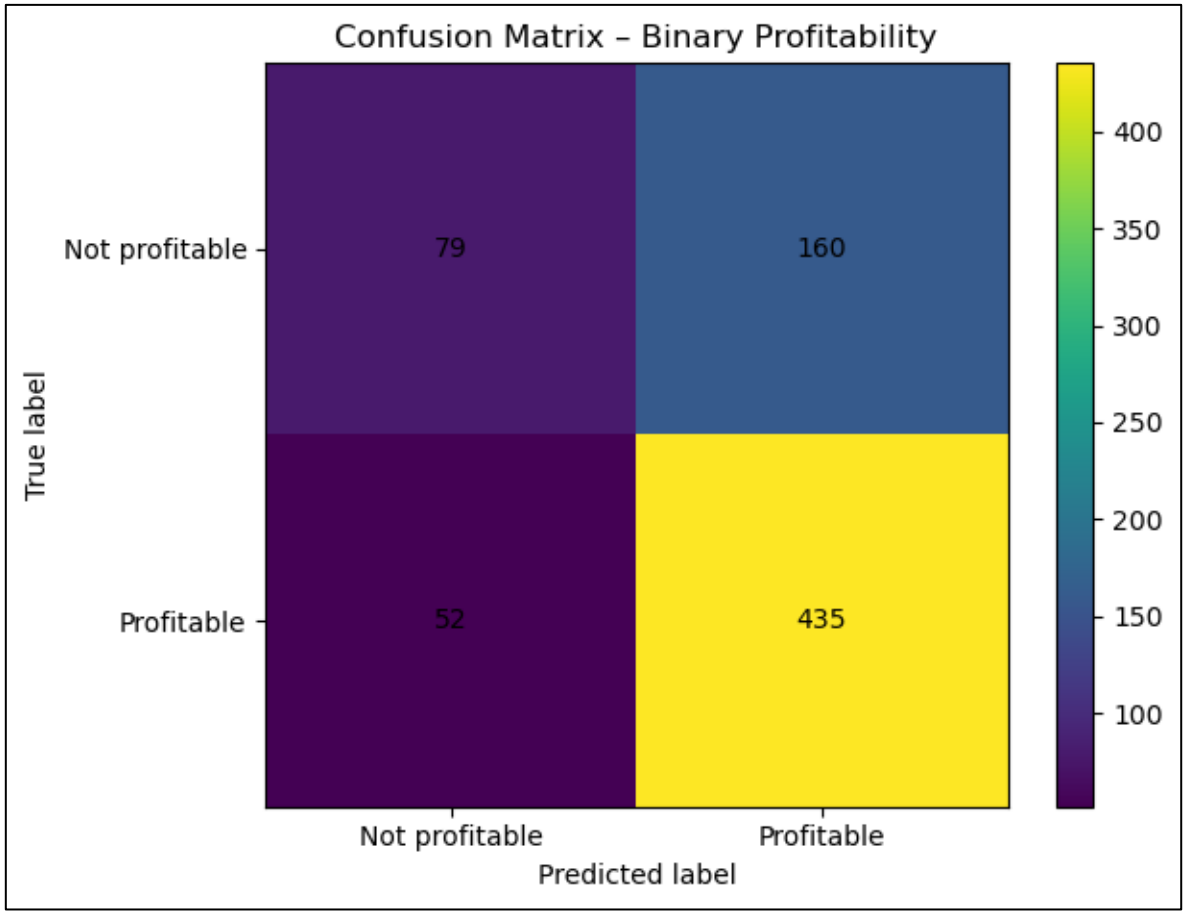


Fig 7. Confusion Matrix

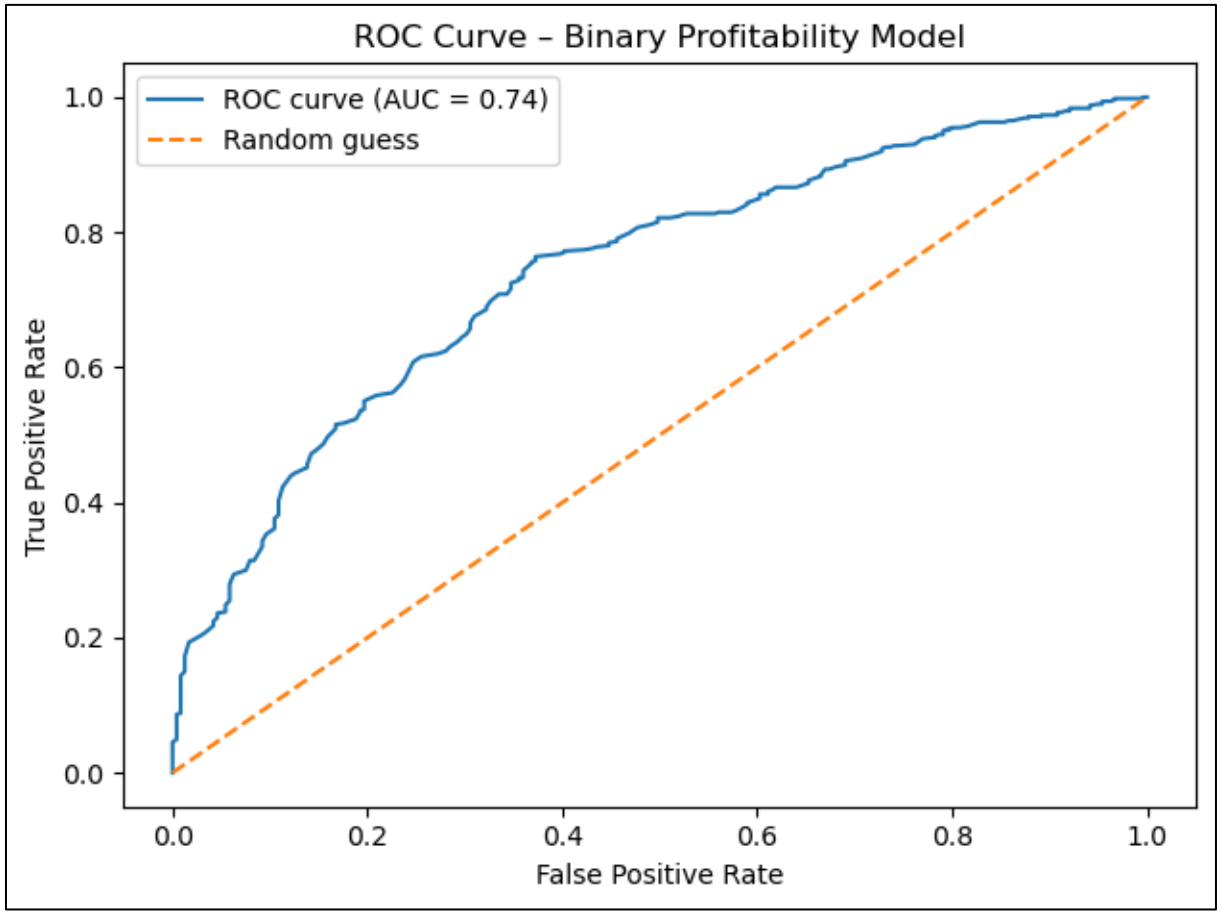


Fig 8. ROC-AUC curve

Exhibits – Supervised Learning

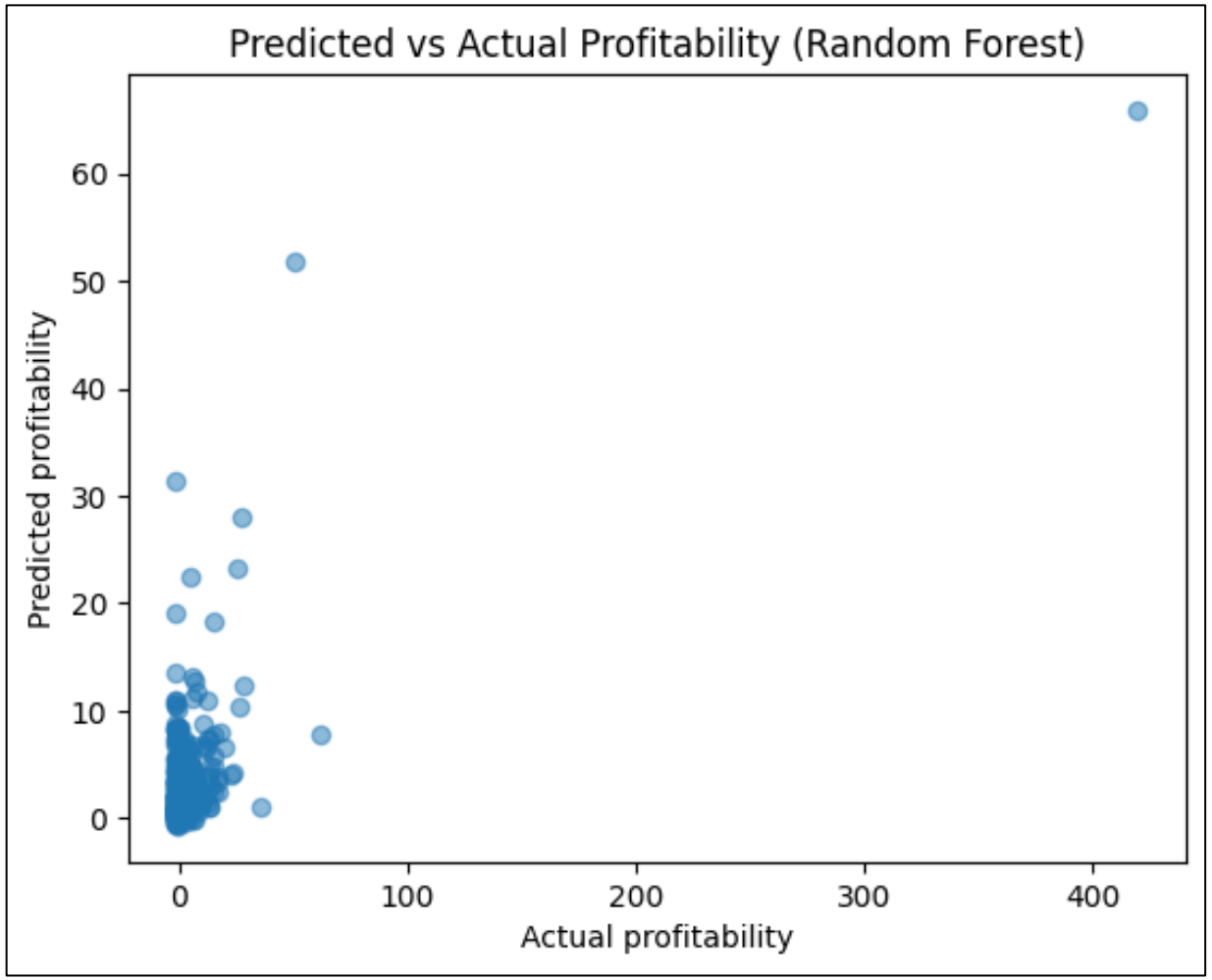


Fig 9. Actual vs Predicted Profitability (Scatter plot)

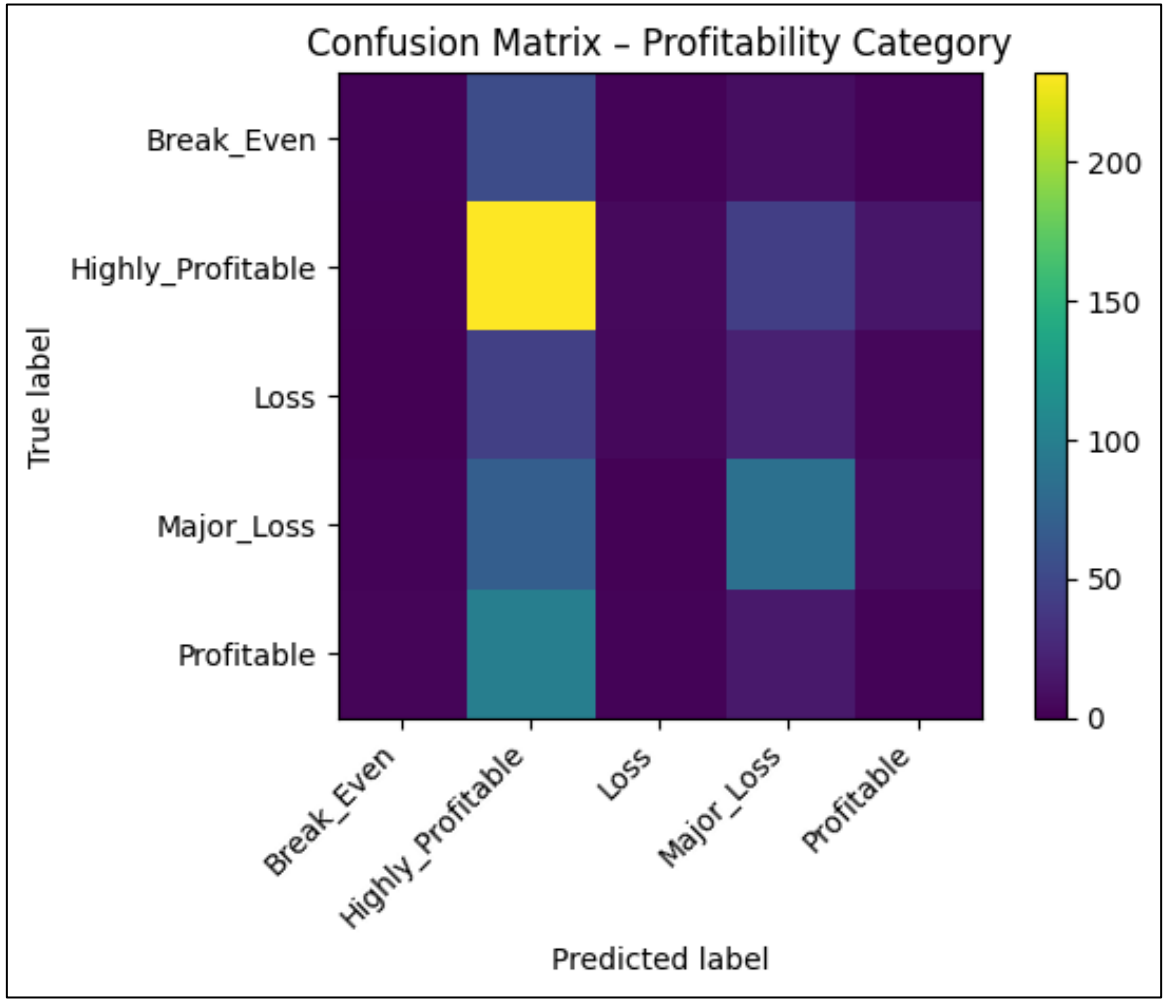


Fig 10. Confusion Matrix (Random Forest model)

Exhibits – Supervised Learning

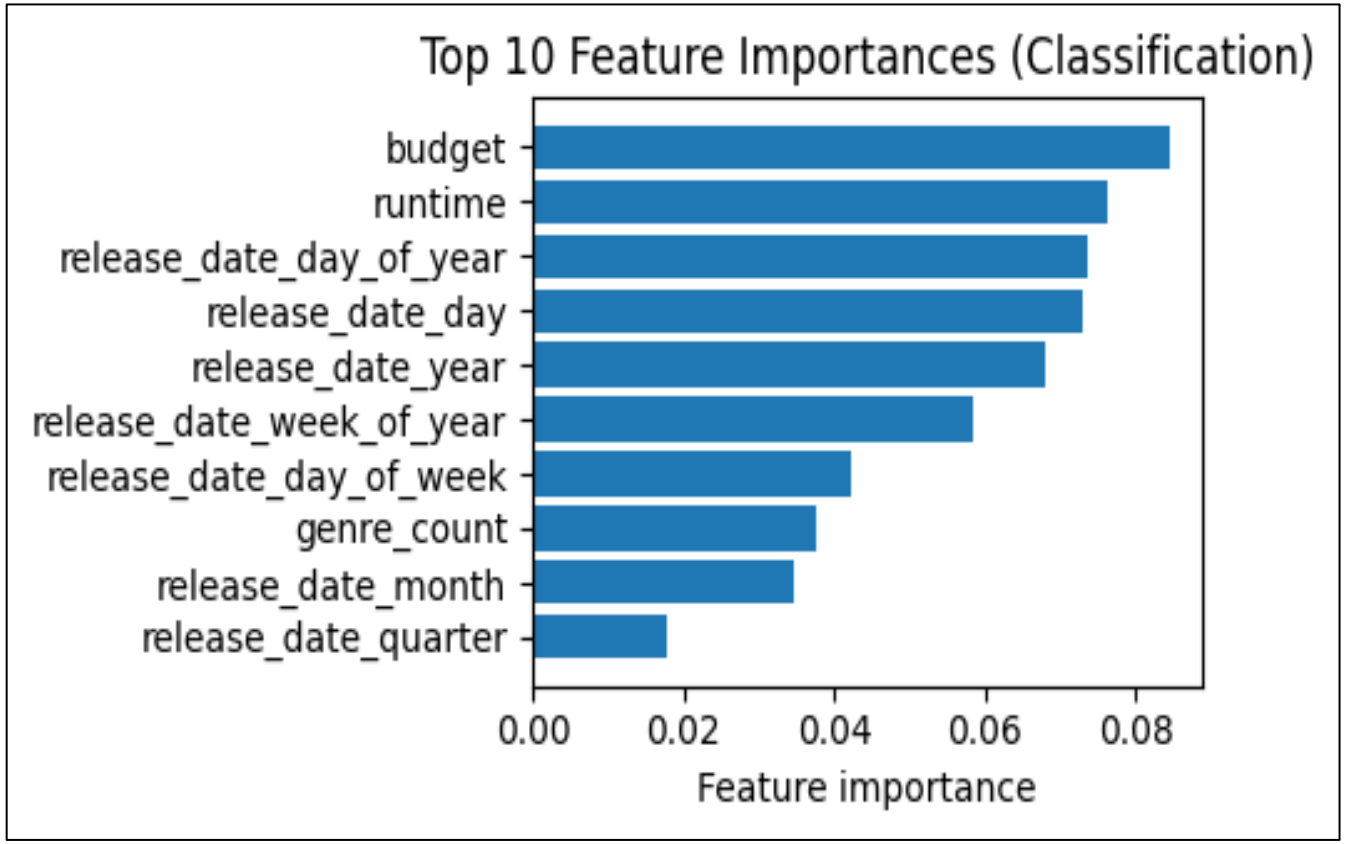


Fig 11. Top 10 important features

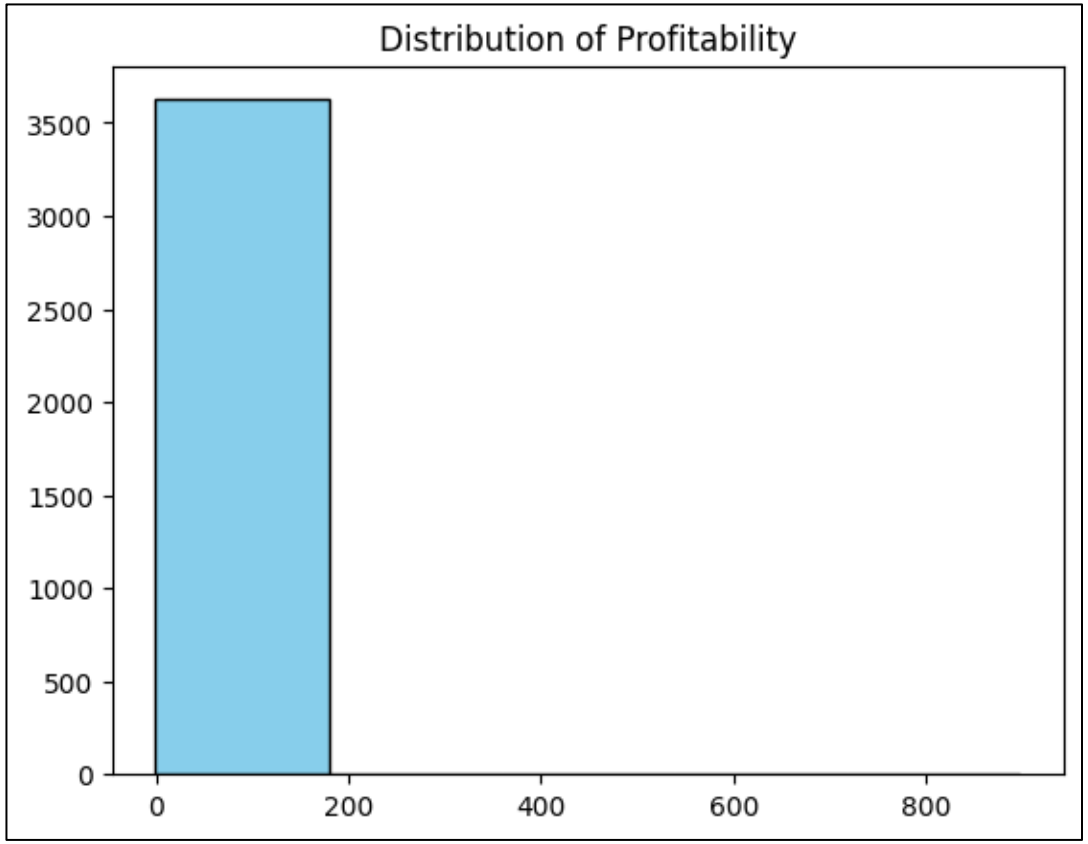


Fig 12. Profitability Distribution